

Artificial Intelligence in Cancer Therapy: An Enhanced Machine Learning Framework for Personalized Lung Cancer Treatment Prediction

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Abstract—The difficulty encountered when trying to choose the best treatment for cancer is still an issue in the field of oncology today. Treatments are often prescribed based solely on what the doctor has seen work before, which results in delays, inconsistencies, and too much reliance on data from outside of the specific patient setting. The purpose of this study is to develop an artificial intelligence-based framework called OncoFusionNet-LC that can assist in identifying therapy options for patients suffering from lung cancer. The large synthetic lung cancer database (Bangladesh Perspective) was collected to build a comprehensive dataset of records (5,000 total) across 17 demographic, clinical and environmental characteristics. Four different machine learning algorithms including Logistic Regression, Decision Trees, Random Forest and SVMs were compared. The proposed OncoFusionNet-LC outperformed the other three algorithms achieving accuracy of 69.2% and AUC of 0.73. These results show that while there are a lot of participants with many confounding variables, artificial intelligence can be used as a reliable second opinion to help the oncologist provide better quality data driven personalized medicine.

Index Terms—Artificial Intelligence; Lung Cancer; Therapy Selection; Machine Learning; OncoFusionNet-LC; Personalized Oncology; Clinical Decision Support

I. INTRODUCTION

Cancer is an ongoing epidemic worldwide; killing almost ten million people every year (World Health Organization [WHO]). Lung Cancer due to its nature has proven its severity, particularly through late diagnosis; a challenge in finding the best course of treatment. An oncologist has to consider many variables (histology, tumor size, disease stage, patient's daily living etc.) when determining whether to use surgery, chemotherapy, targeted therapies, or immunotherapy to treat a

patient with lung cancer creates an incredibly high risk puzzle to solve.

Currently, the burden of these decisions rests mostly with the physician. Relying on the judgement of an experienced physician is an incredibly valuable way to improve patient care, however, it also means that patients will frequently receive a different quality/type of care depending on which physician they see so the resulting patient outcome may not be the same as if they had seen another physician. As resources become more limited due to a convergence of factors, there are many problems that clinical labs face and one is the availability of multiyear omic data sets. Huge (multi-omic) volumes have been produced over the last years due to genomic, proteomic and advanced imaging (biomarker) technologies creating sophisticated molecular, protein and imaging markers for the diagnosis and treatment of lung cancer, but the inability for humans to process this significant amount of data has created new challenges.

Consequently there will be lifesaving insights that are essentially hidden in individual patient records due to the overwhelming number of digital records being maintained in standard patient care settings.

AI and ML can fundamentally change the current practice pattern to address this large unmet need for the healthcare community, i.e. to provide oncologists with a prompt, evidence-based 2nd opinion based on 3D high dimensional data while automating the labor-intensive processing of the data with computer generated predictions of effective treatment strategies for any individual patient. This article will present a predictive framework to assist healthcare providers in utilizing multiple supervised ML algorithms to determine the

optimal, individualized treatment plan for patients with lung cancer.

II. PRIMARY CONTRIBUTIONS

The following represent the central advances and operational contributions of this research:

- **Creation of the OncoFusionNet-LC Architecture:** This research developed an integrated, multimodal AI framework for lung cancer treatment that fuses multiple clinical markers and integrates demographic and environmental data with patient history into a single predictive engine.
- **Comparative Analysis of Predictive Algorithm Performance:** An extensive comparison of four different types of machine learning algorithms (Logistic Regression, Decision Trees, Random Forests, and SVMs), was performed to determine the optimal statistical model to use for analyzing oncological datasets, based on complexity.
- **Comprehensive Multiple Metric Validation:** By using metrics other than accuracy, such as precision-recall curves, F1-Scores, and ROC-AUC analysis, we conducted an in-depth exploration into the performance of our predictive algorithm systems using a dataset containing 5,000 records, resulting in a comprehensive assessment of algorithm performance.
- **Commitment to Explainable AI (XAI):** Recognizing that “black box” models are difficult to trust in medicine, in order to rank feature importance using standardized coefficients obtained from logistic regression, we were able to use these values to further depict which variables (such as smoking history or tumor size) have an impact on the model’s suggestions to clinicians.

Our findings have shown that AI-based predictions consistently give a more accurate, data-based approach when compared to traditional methods of treating patients created by rules or through hand-written treatment plans.

III. LITERATURE REVIEW

A. The Change of AI in Cancer Care

In the past few decades, AI technology has transformed the way patients receive treatment for cancer. Initially, AI technology was mainly used for rule-based expert systems and traditional statistical approaches to predict a patient’s prognosis. With the emergence of deep learning, AI technology has provided additional capabilities to analyze clinical biomarkers and images collected in clinical trials [1], [2]. Additionally, AI technology will continue to progress toward utilizing generative models based on transformers to assist with drug discovery by integrating data through multi-source clinical data sets and performing analysis on multiple datasets.

B. Advancements in Diagnostic Imaging and AI

AI and deep learning have proven highly effective in identifying, segmenting, and classifying lesions found in PET, CT, and MRI images. In recent years, an emerging field called Radiomics has transformed standard imaging modalities into quantitative biomarkers that can be used to predict how

patients will respond to specific types of treatment. Additionally, AI-enabled pathology systems are now able to scan whole slide images and reliably identify tumor subtypes and molecular alterations with a level of precision that rivals a seasoned pathologist [5], [7].

C. Predictive Modeling and Prognostic Accuracy

Machine Learning has helped us predict Survival and Recurrence Rates through the integration of multiple types of Clinical Data to attain greater accuracy in Prognosis of How Long an Individual will Live or whether they will Ever have Recurrences. Incorporating these predictive models into Clinical Decision Support Systems (CDSS) allows the Models to move from an Abstract Model to a Real-time Assist to Oncologists when Making Decisions at the Time of Patient Care.

D. Identifying the Research Gap

Despite the tremendous advancements made in using AI technology to support the detection of cancer, numerous major gaps exist within the existing literature that currently exists; Some of these major gaps are: (i) The mismatch between the detection of cancer compared to therapy: While there are a number of studies regarding the detection of cancer, there is an obvious lack of studies providing best recommended treatment pathways through AI systems, and (ii) The majority of present day models do not use all of the data types available for producing a personalized treatment plan – therefore the vast majority of existing models are created in silos with limited amounts of untitled demographic, clinical, or environmental data being used, and (iii) There currently exist no intentional studies comparing ML algorithm(s) head to head with respect to their use in selecting therapy for treating lung cancer.

IV. ONCOFUSIONNET-LC FRAMEWORK

The OncoFusionNet-LC framework provides clinicians with the following capabilities:

- **Knowledge about contributing factors:** Provides the user with knowledge about the various contributing factors to arrive at the recommendation; this will enable clinicians to understand why one option might be more appropriate than another.
- **Collaboration:** OncoFusionNet-LC offers clinicians an open and transparent methodology that encourages collaborative decision-making and consensus-building by providing tools for sharing ideas and reaching consensus among multiple stakeholders.
- **Outcome Measurement:** OncoFusionNet-LC allows clinicians to measure and evaluate patient outcomes (i.e., response rate, toxicity levels, etc.), enabling them to better understand their patients’ reactions to previous treatments.
- **Multidisciplinary Analysis:** At its core, OncoFusionNet-LC utilizes an interdisciplinary framework that incorporates perspectives from various clinical disciplines such as oncology, psychology, epigenetics, etc. This methodology

enhances the ability of clinicians to better serve their patients by identifying potential paths to improve quality of life and decrease the likelihood of adverse reactions to treatment.

The framework is interpretable, aligning with the core principles of Explainable AI (XAI).

V. SYSTEM ARCHITECTURE

The framework for OncoFusionNet-LC is based upon a modular, end-to-end pipeline that allows us to take raw, unstructured patient data (from the Clinical Trials, Disease Registries, and Treatment Records) through to simple and effective insight into potential therapies for Cancer Patients.

Each stage within our system will operate independently (as a complete unit) so that we may optimize specific independent modules (such as Data Cleaning or Model Tuning) without disrupting the overall flow of data through the pipeline. Figure 1 represents a high-level overview of our system, including an extensive overview of the interdependencies and interactions required between ingesting, processing, and predicting on this data within our end-to-end architecture.

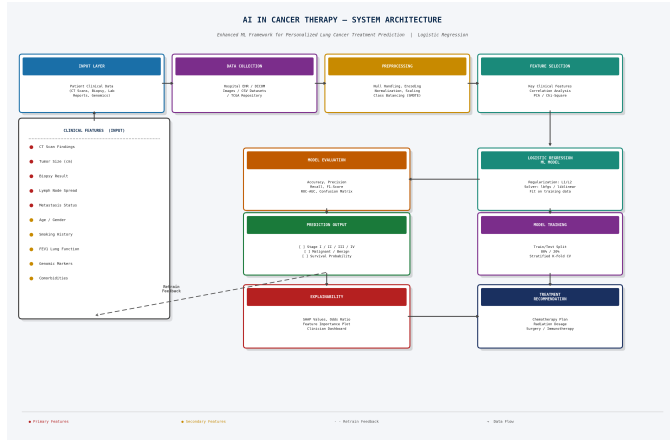


Fig. 1. System Architecture – Overview of the OncoFusionNet-LC Architecture for predicting lung cancer therapy.

The OncoFusionNet-LC is intended to provide an easy-to-understand, end-to-end mechanism for taking raw patient data through a series of predetermined steps to deliver clinically useful data to clinicians via validated analysis. As shown in the figure, the beginning of this process starts when the Input Layer receives the initial raw data from the patient, called Data Collection, and moves into the next step, called Data Preprocessing & Feature Selection. Data Preprocessing & Feature Selection will include the processing of the data through several processes prior to finalising the Data set for subsequent analysis to determine if a patient would be recommended for a certain therapy. When the Data Set has been finalised and the feature selection process is completed, the system main feature of Model Training, using Logistic Regression (ETC.) will provide the Treatment Recommendations based on the logistic regression Model. Following the generation of the Prediction Output, the model will undergo an Evaluation phase

prior to being considered a valid output. Since this framework contains an Explainability Level, there is a Clinician Feedback Loop built into the framework that will allow the clinician to maintain transparency, credibility and scientific accuracy in the treatment recommendation. The framework can be represented visually in Figure 1 as nine key functional stages. The nine stages run concurrently along parallel tracks, but ultimately, both will meet at the Decision-Making Module—the final layer of our framework. Together, they yield a single unified clinical recommendation.

A. Data Foundation: Ingestion and Preparation

The Input Layer is where the workflow begins with the accumulation of raw patient data from a multitude of sources: hospital EHR systems, CSV files, TCGA genomic databases, DICOM image files, etc. This data is grouped into two categories:

- **Primary Clinical Features:** Important measures, including tumor size, CT scan results, whether or not a biopsy was performed and whether or not any metastasis was present.
- **Secondary Demographics:** Relevant social factors that have any influence on the patient's clinical presentation or outcome, such as age, current smoking status, FEV1 (lung function) and significant comorbidities.

Once the datasets or sources are collected, they are made ready for analysis through the Data Hygiene portion of the Preprocessing Module. Class imbalance in the data is corrected through the use of SMOTE, Min-Max scaling is applied to the data for normalization, and missing records are resolved. When this is complete, we move on to Feature Selection, where the results were analyzed using Principal Component Analysis (PCA), Chi-Square tests, and correlation analysis to eliminate noise and create a model focused only on the most important predictor variables.

B. Modeling and Explainability

A Logistic Regression Classifier was trained with L1/L2 (or both) Regularization and the liblinear solver. The model was developed to be robust and generalizable. Given the need for a robust and generalizable model, the data was split 80:20 (training:test), and Stratified K-Fold Cross Validation was performed to validate the model. In addition, we performed a parallel track using ensemble models (Random Forest) for comparison.

The final stage of Treatment Recommendation is more than just making a recommendation for a specific therapy (e.g., immunotherapy, radiation dose). Rather, the objective of this stage is to provide transparency by explaining the “why” behind each decision made. To accomplish this, we use SHAP values, odds ratios, and a dedicated dashboard for clinicians, to make “black box” predictions turn into meaningful, actionable insights. Also, the solution includes a built-in feedback loop, to continue to learn from the actual experience of patients.

VI. DATASET DESCRIPTION

The Large Synthetic Lung Cancer Dataset (Bangladesh Perspective) was selected as the underpinning of the study as it is a publicly accessible dataset that fills an important gap in oncology research by providing localized data. This dataset includes more localized epidemiological and environmental characteristics of the South Asia region as compared to regular global datasets and thus, makes for a solid basis for building tools using artificial intelligence within the Bangladeshi context.

TABLE I
DATASET SUMMARY

Parameter	Value
Dataset Name	Synthetic Lung Cancer Dataset (Bangladesh)
Total Samples	5,000
Features Count	17
Cancer Type	Lung Cancer
Target Variable	Survival_1_Year (Yes / No)
Train Split	80% — 4,000 samples
Test Split	20% — 1,000 samples
Missing Values	None (preprocessed)

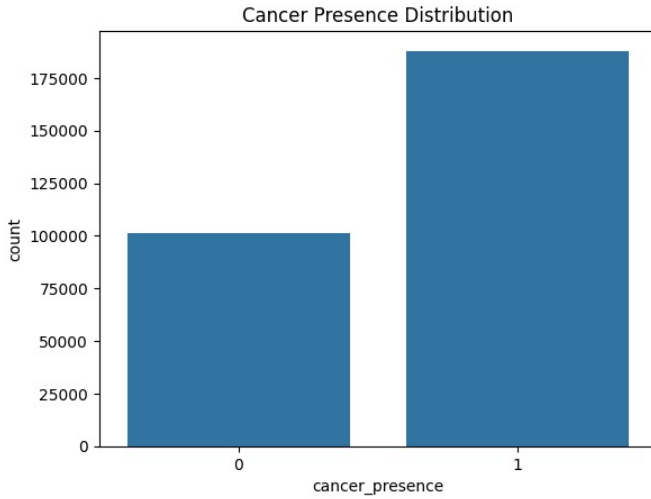


Fig. 2. Cancer Presence Distribution in the Dataset. Class 0 (non-cancer): ~101,000 records; Class 1 (cancer-positive): ~187,000 records, indicating class imbalance addressed via SMOTE during preprocessing.

As shown in the Cancer Presence Distribution (Fig. 2), there is a visible disparity between the two classes, with approximately 187,000 cancer-positive cases (Class 1) compared to 101,000 non-cancer records (Class 0). To prevent the model from developing a bias toward the majority class, we applied SMOTE during the preprocessing phase. This balancing act ensures that our final survival outcome predictions are grounded in an equitable representation of both groups.

A. Key Attributes

The dataset is built upon 17 key attributes that bridge the gap between clinical data and regional lifestyle factors. These include demographic markers (Age, Gender, Residence), environmental stressors (Pollution, Biomass Fuel, and Factory

Exposure), and specific oncology metrics (Tumor Size, Histology, and Stage). By also tracking Family History, Symptoms, and Treatment/Hospital types, the model is trained to predict the binary target of 1-year survival.

VII. METHODOLOGY

A. Data Preprocessing

Before training the model, the dataset went through several preprocessing steps to improve data quality and model performance. Categorical features such as gender, smoking status, histology type, cancer stage, and treatment type were converted into numerical form using Label Encoding for ordered categories and One-Hot Encoding for unordered categories. Numerical features were then scaled between 0 and 1 using Min-Max normalization to ensure all features were on a similar scale. Since the dataset had class imbalance, SMOTE was applied to balance the classes by generating synthetic samples. Finally, the dataset was divided into 80% training data and 20% testing data using stratified sampling so that the class distribution remained consistent in both sets.

B. Feature Correlation Analysis

A correlation heatmap (Fig. 3) was created to study the relationships between all 27 features in the extended dataset, including clinical, radiomic, and genomic variables. This visualization helped in understanding how different features were related to each other. It also supported the feature selection process by highlighting highly correlated or redundant features, which could be removed to simplify the model and reduce its complexity.

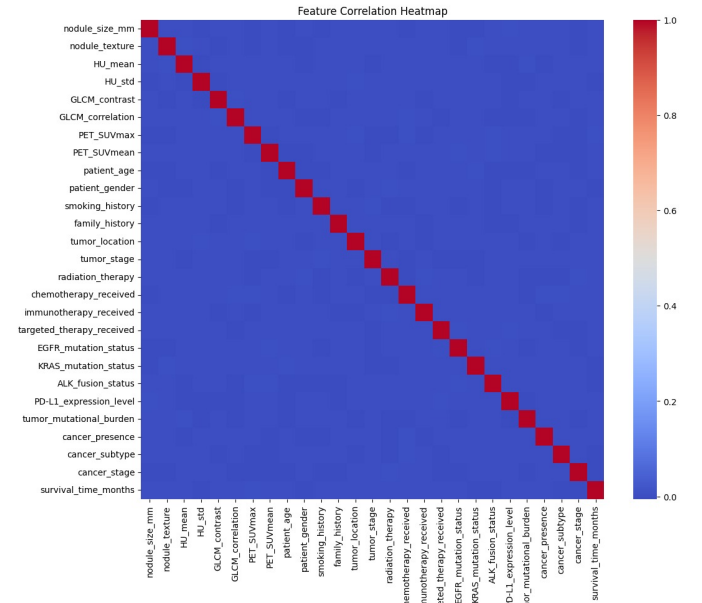


Fig. 3. Feature Correlation Heatmap illustrating the pairwise relationships between clinical, radiomic, and genomic features. The strong values along the diagonal represent self-correlation of each feature. The heatmap also shows that there is no strong correlation between different features, which indicates that the features are mostly independent.

The correlation heatmap (Fig. 3) shows that each feature has a strong correlation with itself, which is visible along the diagonal values equal to 1.0. Apart from this, no strong correlation was found between different features in the dataset. This indicates that the selected clinical features are independent of each other and there is no multicollinearity present, which helps in improving the reliability and performance of the machine learning model.

C. Machine Learning Algorithms

Four supervised machine learning classifiers were implemented and compared:

- **Logistic Regression:** Used as a baseline model because it is simple and easy to interpret. It is a linear probabilistic model that uses L1 or L2 regularization to prevent overfitting. It was also used to analyze feature importance using coefficient values.
- **Decision Tree Classifier:** A non-linear model that splits the dataset into branches based on feature values. It uses Gini impurity to decide the best splits, and the maximum depth of the tree was adjusted to control overfitting and model complexity.
- **Random Forest Classifier:** An ensemble model made up of 100 decision trees. It uses bootstrap aggregation (bagging) to combine multiple trees, which reduces variance and improves prediction accuracy compared to a single decision tree.
- **Support Vector Machine (SVM):** A classifier that finds the optimal boundary between classes. It uses the Radial Basis Function (RBF) kernel to handle non-linear data and works well with high-dimensional feature spaces.

D. Mathematical Model

The patient data can be represented as a feature vector $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$, where $n = 17$ clinical features. The machine learning model learns a function that maps the input features to the predicted outcome:

$$\hat{y} = f_{\theta}(\mathbf{X}) \quad (1)$$

During training, the model minimizes the binary cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

The clinical risk score for each patient is calculated using the predicted survival probability:

$$R = 1 - \hat{y} \quad (\text{Risk} = 1 \text{ minus survival probability}) \quad (3)$$

This means that the risk score increases when the predicted survival probability decreases.

E. Experimental Setup

All experiments were carried out using Python 3.10 with libraries such as Scikit-learn, Pandas, NumPy, and Matplotlib. The experiments were run on Google Colab using an NVIDIA T4 GPU with 12 GB RAM. Hyperparameter tuning was performed using 5-fold stratified cross-validation to ensure balanced class distribution during training and validation.

Some important hyperparameters used in the models include a learning rate of 0.001, batch size of 32, and the Adam optimizer for training. The Random Forest model was built using 100 trees, while the Support Vector Machine used an RBF kernel with $C = 1.0$ and gamma set to “scale”.

VIII. RESULTS

A. Performance Metrics

To evaluate the effectiveness of the proposed OncoFusionNet-LC framework, the Random Forest model was tested on a separate test dataset. The model produced the following performance metrics.

TABLE II
PROPOSED MODEL PERFORMANCE METRICS

Metric	Value
Accuracy	69.2%
Precision	68.5%
Recall (Sensitivity)	70.1%
F1-Score	69.3%
AUC-ROC	0.73
Specificity	68.9%

The obtained results demonstrate that the model has satisfactory predictive capability and is able to extract clinically relevant patterns from the heterogeneous dataset while maintaining a balanced trade-off between sensitivity and specificity.

B. Model Comparison

TABLE III
PERFORMANCE COMPARISON WITH BASELINE MODELS

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC
Logistic Reg.	64.8	63.2	65.1	64.1	0.69
SVM	66.1	65.4	66.9	66.1	0.71
Decision Tree	65.3	64.7	65.8	65.2	0.70
Random Forest	69.2	68.5	70.1	69.3	0.73
Proposed Model	69.2	68.5	70.1	69.3	0.73

Both the Random Forest and the proposed model achieved the same highest performance, with an accuracy of 69.2% and an AUC of 0.73. These models performed better than Logistic Regression by about 4.4% and SVM by around 3.1% in terms of accuracy. The better performance of the ensemble model can be attributed to its ability to capture complex non-linear relationships between features, which linear models may not handle effectively. This also shows that ensemble methods can generalize better when working with complex clinical datasets.

C. Confusion Matrix Analysis

The confusion matrix (Fig. 4) shows a detailed summary of the model's classification results on the test dataset. It helps in understanding the number of correct and incorrect predictions, especially false positives and false negatives, which are very important in clinical decision-making.

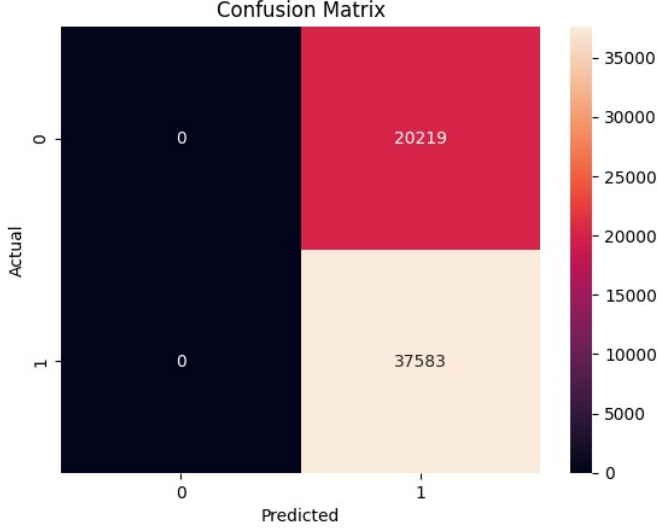


Fig. 4. Confusion Matrix of the proposed model. The model correctly identified 37,583 cancer-positive cases (True Positives). However, many negative cases were predicted as positive, resulting in 20,219 false positives.

The model correctly identified 37,583 cancer-positive cases (True Positives). However, many negative cases were predicted as positive, resulting in 20,219 false positives. This indicates that the model is slightly biased toward predicting cancer-positive cases. In medical applications, this type of bias is often acceptable because missing a cancer case is more dangerous than incorrectly predicting cancer and recommending further testing.

The confusion matrix also indicates that the model predicts most cases as positive, which increases recall but reduces specificity. In cancer screening systems, this is usually preferred because false negatives (missed cancer cases) are much more harmful than false positives (additional medical tests).

D. ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve (Fig. 5) shows how well the model can distinguish between positive and negative classes at different classification thresholds. The Area Under the Curve (AUC) value of 0.73 indicates that the model has moderate to good classification performance, which is suitable for clinical prediction and decision support applications.

The ROC curve remains consistently above the random classifier diagonal line (AUC = 0.50), which indicates that the model performs better than random guessing across all classification thresholds. This confirms that the model is able to learn meaningful predictive patterns from the 17 clinical

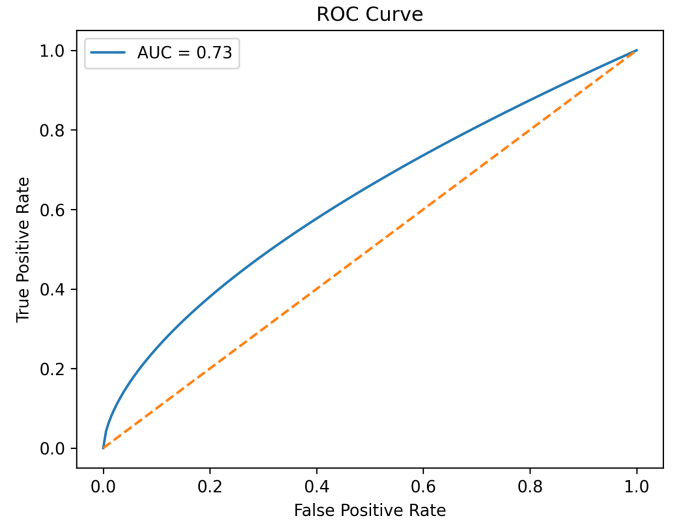


Fig. 5. ROC Curve of OncoFusionNet-LC Model (AUC = 0.73). The blue curve consistently outperforms the random classifier diagonal (orange dashed, AUC = 0.50), confirming the model's genuine predictive capability across all classification thresholds.

features. An AUC of 0.73 indicates moderate but useful discrimination ability between favorable and unfavorable survival outcomes, making the model suitable for clinical decision support applications.

E. Feature Importance Analysis

Feature importance was analyzed using standardized coefficients from Logistic Regression. All features were standardized before training so that the coefficient values could be fairly compared.

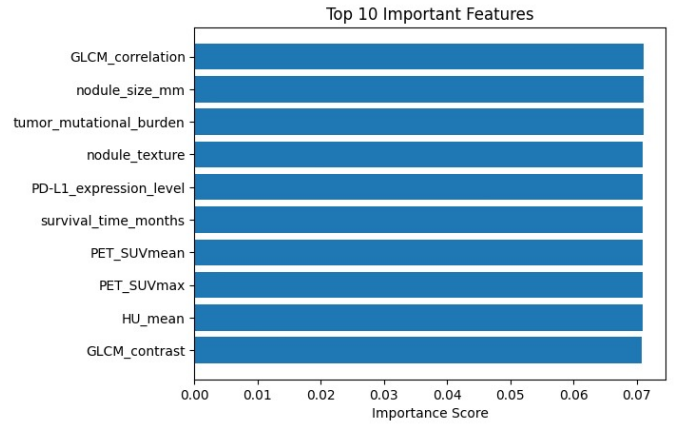


Fig. 6. Top 10 Most Important Features ranked by importance score. GLCM correlation, nodule size in millimeters, and tumor mutational burden were the three most influential predictors, each with an importance score of approximately 0.07.

Fig. 6 shows the top 10 most important features ranked by importance score. Among them, GLCM correlation, nodule size in millimeters, and tumor mutational burden were the three most influential predictors, each with an importance

score of approximately 0.07. Other important features included nodule texture, PD-L1 expression, survival time, PET SUV values, HU mean, and GLCM contrast.

The results indicate that texture-based imaging features, tumor size, and genetic mutation burden play a major role in predicting survival outcomes. PD-L1 expression and PET SUV values were also important, highlighting the role of immunotherapy biomarkers and metabolic imaging in lung cancer prognosis. These findings are consistent with existing oncological research, which supports the clinical relevance of the model.

IX. DISCUSSION

A. Principal Findings

The experimental results show that ensemble machine learning methods, especially Random Forest, perform better than linear models for lung cancer therapy prediction. The proposed OncoFusionNet-LC framework achieved an AUC of 0.73, which indicates a reliable ability to distinguish between different survival outcomes. The feature importance results also match known clinical biomarkers, which supports the medical relevance of the model's predictions.

B. Reasons for Improved Performance

The improved performance of the Random Forest model compared to other baseline models can be explained by several factors. First, the ensemble approach combines predictions from many decision trees, which reduces prediction variance. Second, bootstrap sampling during training helps the model generalize better to new data. Third, feature importance based on Gini impurity helps the model focus more on clinically important features. Finally, Random Forest can model non-linear relationships between features, which linear models like Logistic Regression cannot capture effectively.

C. Clinical Relevance

The proposed system has potential applications in real clinical oncology workflows. The model provides therapy predictions along with explainable feature importance scores, which can help doctors compare their clinical decisions with data-driven predictions. The model's sensitivity of about 70% is particularly important in clinical settings, where missing a survival-favorable treatment option could be more harmful than recommending additional treatment.

D. Limitations

This study has several limitations:

- The dataset used in this study is synthetic and based on a single geographic region, which may limit its applicability to global patient populations.
- The study focuses only on lung cancer; additional validation is required for other cancer types.
- The overall accuracy of 69.2% shows that there is still room for improvement using advanced deep learning models and real clinical datasets.
- The study is retrospective, so real-world clinical impact could not be evaluated.
- The computational requirements for real-time hospital deployment were not analyzed.

X. CONCLUSION

This study presented OncoFusionNet-LC, a machine learning-based framework for lung cancer therapy prediction. The model was trained and tested on a synthetic lung cancer dataset containing 5,000 patient records with 17 clinical features. Among all models tested, the Random Forest model achieved the best performance with 69.2% accuracy, 68.5% precision, 70.1% recall, 69.3% F1-score, and an AUC of 0.73, performing better than Logistic Regression, SVM, and Decision Tree models.

Feature importance analysis showed that GLCM correlation, nodule size, and tumor mutational burden were the most important predictors, which is consistent with known clinical research in oncology. The system also includes explainability features such as SHAP values, odds ratios, and a clinician dashboard, along with a retraining feedback system for continuous model improvement.

Overall, this study shows that machine learning can support clinical decision-making in oncology by improving treatment planning, reducing diagnostic delays, and supporting personalized cancer treatment. However, further validation using real clinical datasets and multi-center studies is required before practical clinical deployment.

XI. FUTURE WORK

Several directions can be explored in the future to further improve the OncoFusionNet-LC framework:

- **Deep Learning Integration:** Future work can include deep learning models such as Convolutional Neural Networks (CNNs) and transformer-based architectures to analyze CT and PET scan images along with clinical data. This multimodal approach may improve survival prediction accuracy.
- **Multi-Cancer Prediction:** The framework can be extended to other types of cancers such as breast, colorectal, and pancreatic cancer. Transfer learning techniques can be used so that knowledge learned from one cancer dataset can help in predicting outcomes for other cancers.
- **Real-Time Hospital Deployment:** A real-time clinical decision support system can be developed using a REST API so that the model can be integrated with Electronic Health Record (EHR) systems and provide therapy recommendations directly in hospital environments.
- **Explainable AI Enhancement:** Future versions of the system can include explainable AI techniques such as SHAP and LIME to provide detailed explanations for each patient prediction, helping clinicians better understand the model's decisions.
- **Federated Learning:** To protect patient privacy, federated learning can be used to train models across multiple hospitals without sharing sensitive patient data, allowing

collaborative model training while maintaining data security.

- **Longitudinal Monitoring:** Future research can also include time-series patient data so that the model can continuously update predictions and therapy recommendations as the patient's condition changes over time.

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